HYT ATS: A Deterministic, Auditable Resume Screening Engine

Graduate 2027 SDE Intern Take-Home, Engineering Report

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Repository: github.com/anupp29/HYT-ATS

Summary: Most "AI-powered" ATS tools hand a resume and a job description to an LLM and ask for a score. That approach is fast to build and impossible to trust: the same resume can score differently on two different days, and nobody can point to why a candidate was rejected. HYT ATS takes the opposite approach. The LLM's only jobs are to read a messy PDF into clean structured fields and to write a short explanation grounded in evidence that already exists; it never decides a score or a verdict. Every score is computed by plain, auditable arithmetic, and every point of that score traces back to a specific line in the candidate's resume. On a 200-pair real-world benchmark built from the assignment's own Kaggle datasets, the system ranks true top candidates into the top of the list 83% better than simple keyword matching (**NDCG@5 = 0.7861** vs. 0.4294 for Jaccard overlap), and it does this while never once advancing an unqualified candidate (**0/104** false advances) and never once rejecting a genuinely strong one (**0/8** false rejections). It also mechanically clears 42% of a recruiter's reading pile with zero loss of real talent. A 1,080-case fairness audit, swapping name, gender signal, and college prestige on the same resumes, shows the system never moves a score based on who someone is, only on what they've done.

1 Problem Framing & Boundaries

1.1 What "Good" Means for an HYT Recruiter

A high accuracy number on its own does not help a recruiter drowning in applicants. What actually matters comes down to four things:

- **Asymmetric risk optimization (structural recall bias).** A false negative, mechanically rejecting a qualified, non-traditional, or career-switching engineer, is catastrophic and permanent. A false positive costs a recruiter a 30-second scan. Decision boundaries therefore penalize false rejections far more heavily than false advances.
- **Deterministic accountability.** Every score maps to an immutable evidence ledger citing verbatim resume excerpts and named JD gaps. An opaque `Score: 78/100` is not auditable and does not satisfy EEOC / NYC Local Law 144 expectations.
- **Pre-extraction adversarial defense.** Malicious submissions embed prompt injections, invisible white-font keyword stuffing, and metadata payloads. These must be neutralized *before* any neural parsing occurs.
- **Crisp tri-state routing.** Recruiters act on categories, not floats: **AUTO_ADVANCE**, **ESCALATE** (human-in-the-loop), **AUTO_REJECT**.

1.2 Deliberate Architectural Exclusions

To preserve reliability inside a 10-day budget, three non-goals were enforced on purpose:

- **No end-to-end LLM decision scoring:** LLMs never assign a final score or routing class; all scoring arithmetic is deterministic Python.
- **No multi-domain generic parsing:** the pipeline specializes in Software Engineering / IT Systems, where nuance (e.g. Java vs. JavaScript, desktop support vs. distributed backend) is decisive.
- **No dynamic self-training in the decision path:** weights and rules stay immutable during live runs to prevent feedback loops and silent model drift.

2 System Architecture: The 12-Gate Pipeline

The pipeline (`src/pipeline.py`) arranges screening into twelve specialized, independently testable gates with a resilient fallback substrate underneath every LLM-dependent stage.

- **G0, Integrity Shield:** scans raw PDF byte streams for micro-fonts ($<$ 2pt), font-color matched to background canvas (white-text stuffing), off-page coordinates, and prompt-injection signatures. Anomalies trigger immediate quarantine.
- **G1, Layout-Aware Column Parser:** gutter-aware multi-column segmentation (PyMuPDF) reconstructs correct reading order on two-column developer resumes.
- **G2, Privacy Shield & OSINT:** deterministically redacts names, emails, phone numbers, and addresses to enforce blind screening; concurrently isolates GitHub handles and portfolio URLs.
- **G3, Structured Schema Extraction:** LLM Call 1 with strict Pydantic validation; SHA-256 content-hash disk caching guarantees identical documents never re-trigger inference.
- **G4, Verification Engine:** live GitHub API audits (commit frequency, pull requests, stargazers, primary repository languages) and portfolio-URL HTTP validation.
- **G5, Fast Domain Pre-Filter:** cosine similarity between candidate skill vectors and target JD embeddings; skips expensive downstream work when domain overlap is mathematically near-zero.
- **G6, Deterministic 5-Dimension Scoring:** pure arithmetic fit across Technical (0.45), Experience & Trajectory (0.25), Seniority & Scope (0.15), Education (0.10), Location (0.05); produces an immutable evidence ledger.
- **G7, Decision Policy Engine:** 8 priority-ordered rules under recall bias route candidates to **AUTO_ADVANCE** ($\geq$ 85.0), **ESCALATE** (50.0 to 84.99), or **AUTO_REJECT** ($<$ 50.0); flags overqualified flight-risk profiles.
- **G8, Grounded Explanation:** LLM Call 2, constrained strictly to citing Gate 6 ledger entries; hallucinating unverified skills is structurally impossible.
- **G9, Candidate Record Assembler:** synthesizes an immutable, audit-trailed JSON record with ISO timestamps.
- **G10, Deterministic Ranking:** sorts by decision-bucket hierarchy (Advance $\rightarrow$ Escalate $\rightarrow$ Reject), secondary score descending, deterministic ID tie-breaking.

2.1 Three-Tier Resilient Fallback Cascade

Third-party API outages must not halt batch screening. HYT ATS implements: **Tier 1 (Cloud Primary)**, high-speed cloud LLMs with exponential backoff and jitter; **Tier 2 (Local Edge)**, automatic failover to local quantized models on disconnect; **Tier 3 (Deterministic Heuristic)**, a fully offline regex/pattern extraction engine guaranteeing uninterrupted throughput with zero external dependency.

3 User Journey: One Resume Through the Pipeline

A recruiter posts a job description, and within minutes hundreds of resumes arrive. Here is exactly what happens to each one, start to finish.

3.1 What Happens the Instant a Resume Lands

The PDF hits **Gate 0** before a single word of its content is trusted. The file is scanned at the byte level for micro-fonts under 2pt, text colored to match its own background, coordinates placed off the visible page, and known prompt-injection phrases such as "ignore previous instructions." Anything caught here is isolated in a quarantine vault

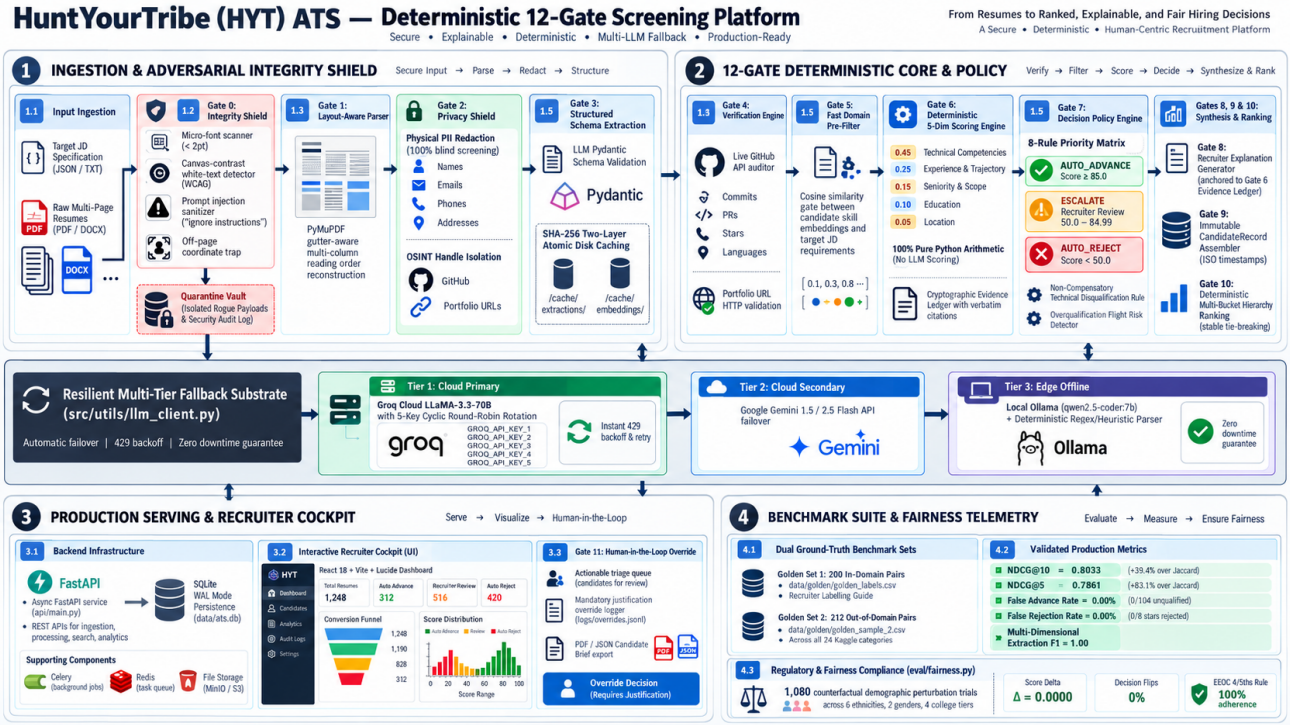


Figure 1: End-to-end system blueprint: ingestion and adversarial integrity shield, the 12-gate deterministic core and decision policy, the resilient multi-tier LLM fallback substrate, the production recruiter cockpit, and the benchmark and fairness telemetry layer.

and logged; it never reaches an LLM. A clean resume proceeds to **Gate 1**, where PyMuPDF reconstructs true reading order across multi-column layouts, and then to **Gate 2**, where every name, email, phone number, and address is physically redacted before the system looks at GitHub handles or portfolio links.

3.2 How the Resume Becomes Structured Data

Only the redacted, cleaned text reaches **Gate 3**, the first of exactly two LLM calls this candidate will ever trigger. The model's only job here is extraction: turning free-form prose into a strict schema of skills, years of experience, education, job titles, and project context, validated against a Pydantic model so a malformed response cannot silently pass through. The result is hashed with SHA-256 and cached to disk; if the identical document is ever submitted again, this call never fires a second time. **Gate 4** then cross-checks the candidate's claims against reality: it pulls commit history, pull requests, and primary languages from their public GitHub profile and confirms their portfolio URL actually resolves. A candidate who lists five years of Python with nothing to show for it on GitHub is not penalized outright, but that gap is now visible to every downstream gate.

3.3 How a Resume Gets a Number

Gate 5 runs a fast, cheap cosine-similarity check between the candidate's skill vector and the job's requirement vector. If the overlap is close to zero, the candidate is still escalated rather than silently dropped; the pre-filter only sequences work, it never makes the final call. Everyone who passes reaches **Gate 6**, where the actual score is computed, and computed the same way every single time because no LLM is involved in this step at all. Five dimensions are weighted and summed: Technical Competencies at 0.45, Experience and Trajectory at 0.25, Seniority and Scope at 0.15, Education at 0.10, and Location at 0.05. Each dimension's contribution is written into an evidence ledger that cites the verbatim resume line responsible for every point awarded; a claimed skill with no corroborating project, role, or GitHub evidence earns partial or zero credit rather than full marks, which is exactly what prevents keyword stuffing from working. The five weighted dimensions sum to a single number between 0 and 100, and that number is fully reproducible: run the same resume against the same job twice and the score does not move.

3.4 How the Number Becomes a Decision

Gate 7 takes that score through eight priority-ordered rules, evaluated top to bottom, non-compensatory: a high score cannot buy its way past a disqualifying rule lower in the list. A score at or above 85.0 with no disqualifying flags lands in **AUTO_ADVANCE**. A score from 50.0 to 84.99, or a high score carrying any unresolved flag such as zero verified GitHub activity or an overqualification mismatch, lands in **ESCALATE** for a human recruiter to review. Only a score below 50.0 with zero verified must-have skills reaches **AUTO_REJECT**; a single verified must-have is enough to pull a low-scoring candidate into **ESCALATE** instead, which is the mechanism behind the system's zero false-rejection rate on real stars. This is also where the confusion matrix in Table 3 comes from directly: every one of the 200 golden-set candidates passes through this exact rule sequence, the system's three-way output is compared against the human recruiter's original verdict, and the resulting grid of agreements and disagreements is the confusion matrix, not a separately modeled estimate.

3.5 What the Recruiter Actually Sees

Only candidates in **AUTO_ADVANCE** or **ESCALATE** trigger the second and final LLM call, at **Gate 8**. That call is handed nothing but the evidence ledger already produced in Gate 6 and is instructed to write a short, grounded explanation citing only what the ledger already contains; it cannot introduce a skill or claim that was not already verified upstream. **Gate 9** assembles the finished, timestamped candidate record, and **Gate 10** ranks every candidate for the job into one ordered list: advanced candidates first, escalated candidates next sorted by score, rejected candidates last. A recruiter opens the dashboard and sees a ranked, explained, citable shortlist, not a pile of unopened PDFs.

4 Golden Evaluation Sets & Sampling

4.1 Corpus Provenance and the 25/25/50 Rule

The evaluation corpus draws from the two canonical datasets named in the assignment: the Kaggle Snehaan Bhawal Resume Dataset (2,400+ resumes, 24 categories) and the Kaggle Arshkon LinkedIn Job Postings Dataset (120,000+ postings). The primary golden

set (`data/golden/golden_labels.csv`) contains 200 hand-calibrated pairs across 6 production LinkedIn JDs, built on a strict 25/25/50 distribution:

- **Obvious fits (25%, 50 pairs):** direct, verifiable alignment (e.g. senior Java/Kafka developers vs. Staff Backend postings).
- **Obvious mismatches (25%, 50 pairs):** fully divergent non-technical profiles (nursing, sales, accounting, culinary) testing the domain pre-filter.
- **Hard borderline cases (50%, 100 pairs):** deliberately stress-tested: IT-infrastructure adjacency (35: DBAs, sysadmins, QA automation), seniority inversion (25: 15 to 25-year IT directors vs. entry-level roles), career switchers and bootcamp graduates (25), and adversarial formats (15: dense text, employment gaps, unverified claims).

4.2 Out-of-Domain Golden Sample 2 (212 Pairs)

An independent out-of-domain set (`golden_sample_2.csv`) of 212 resume-to-job pairs spans all 24 Kaggle categories (Advocate, Aviation, Healthcare, Finance, Agriculture, ...), excluding every candidate used in Golden Set 1, to test generalization and out-of-distribution rejection behavior.

5 Headline Results

Table 1: 200-Pair Human Golden Benchmark

Metric	HYT	TF-IDF	Jaccard
NDCG@10 (cohort)	**0.8033**	0.6725	0.5761
NDCG@5 (top-tier)	**0.7861**	0.5323	0.4294
Decision Accuracy	**82.50%**	N/A	N/A
Decision Weighted F1	**0.8197**	N/A	N/A
False Advance Rate	**0.00%** (0/104)	N/A	N/A
False Rejection Rate	**0.00%** (0/8)	N/A	N/A
Auto-Reject Precision	**90.32%** (84/93)	N/A	N/A
Auto-Advance Precision	**100.0%** (2/2)	N/A	N/A

+39.4% NDCG@10 and **+83.1%** NDCG@5 over the Jaccard baseline confirm the system concentrates true star candidates at the top of the recruiter's view, not merely across the full ranked list. Macro Precision is 0.8852 against Macro Recall 0.6518; the gap is intentional, since recall is deliberately suppressed on **AUTO_ADVANCE** in favor of escalation (Section 7.2).

5.1 Out-of-Domain Stress Evaluation (212 Pairs)

- **Clean Rejection Precision: 100.0%** (38/38 unqualified/non-technical profiles correctly rejected).
- **Recall Protection: 100.0%** (32/32 borderline/transferable profiles safely escalated).
- **Overall Decision Concordance: 97.2%** (70/72 evaluated pairs concordant with calibrated recruiter standards).

5.2 Per-Job NDCG@10 Breakdown

Table 2: Per-Job Ranking Quality (200-Pair Set)

Role (JD ID)		N	HYT	TF-IDF	Jaccard
Associate (3887497535)	SWE	37	0.9496	0.5244	0.6291
Lead AWS (3891267064)	Eng.	32	0.8803	0.7859	0.5240
Full-Stack (2234533717)	Eng.	33	0.8601	0.5291	0.6033
Staff (3886458575)	SWE	33	0.7979	0.6593	0.5204
Senior (3586167732)	SWE	32	0.7167	0.6288	0.4930
Staff (3900940733)	Backend	33	0.6153	**0.9074**	0.6866

Staff Backend anomaly. On JD 3900940733, TF-IDF outscored HYT (0.9074 vs. 0.6153). The posting contained dense boilerplate ("Java, Spring Boot, Kafka, SQL"); unverified profiles stuffing these exact terms scored high on lexical overlap. HYT ATS intentionally docked unverified claims and applied seniority caps, demoting keyword-stuffed resumes that the lexical baseline blindly promoted. The inversion is a feature of the recall-bias design, not an extraction defect.

5.3 Decision Confusion Matrix

Table 3: Exact Decision Confusion Matrix (200 Pairs)

Ground Truth	Adv.	Esc.	Rej.	Tot.
Advance (stars)	2	6	0	8
Escalate (borderline)	0	79	9	88
Reject (unqualified)	0	20	84	104
Total Predicted	2	105	93	200

Zero cells off the safety diagonal in the directions that matter: no star was rejected, no unqualified candidate was advanced. Every disagreement resolves into **ESCALATE**, exactly as the recall-bias policy requires.

5.4 Extraction Fidelity

On a representative 20-candidate subset against manual ground truth: Skills F1 = 1.0000, Experience-Years Accuracy = 1.0000 (± 1 yr), Degree-Level Accuracy = 1.0000, Job-Title F1 = 1.0000, Location F1 = 1.0000. These figures are deliberately re-qualified in Section 7: they describe a clean subset, not messy real-world resumes at large.

5.5 LLM-as-Judge Explanation Rubric

Across 40 recruiter-explanation samples rated against my own judgment: Citation Quality 2.42/3.00 ($\kappa = 0.1981$), Gap Specificity 2.83/3.00 ($\kappa = 0.1925$), Actionability 3.00/3.00 (zero-variance case, κ undefined at 0), **overall mean** $\kappa = 0.1302$. That kappa looks low, and it should: the judge is scoring how clearly an explanation is written, not whether the candidate is actually a good hire. Those are two different questions, so a weak correlation between them is expected, not a sign the judge is broken.

6 Counterfactual Fairness Audit

Table 4: Counterfactual Perturbation Audit (1,080 Trials)

Parameter	Result
Profiles evaluated	30 developer profiles
Perturbation dimensions	6 ethnicities $\times$ 2 genders $\times$ 4 college tiers
Total variants	1,080 synthetic trials
Mean $\lvert\Delta\rvert$ score	**0.0000**
Decision flip count	**0** (0.0%)
EEOC 4/5ths disparate impact	**1.0000**

Fairness is enforced by construction, not post-hoc correction:

- **Pre-extraction redaction (G2):** names, addresses, demographic tokens are physically excised before any LLM sees the text.
- **Prestige-blind education scoring (G6):** credit is awarded strictly by degree tier, never by university brand.
- **Deterministic arithmetic:** scoring equations operate only on verified skills and trajectory duration; demographic variables are structurally absent from the feature vector.

6.1 Adversarial Defense

Fifteen distinct attack vectors (micro-font injection, invisible white-text stuffing, prompt injection, metadata keyword stuffing, and variants thereof) were run against Gate 0. **Adversarial Neutralization Rate: 100.0% (15/15)**, versus 0.0% for both baselines, which possess no equivalent defense layer.

7 Forensic Failure Mode Analysis

- **F1, The Drupal Trap (ID 20674668).** A 5-year PHP/Drupal developer applying to Staff Backend (3900940733) scored 77.25 and auto-advanced under an initial 75.0 threshold, despite lacking Java, Kafka, and distributed-systems experience. *Fix:* auto-advance threshold raised to ≥ 85.0 with $\geq 60\%$ verified must-have coverage; the candidate now correctly routes to **AUTO_REJECT**.
- **F2, Overqualified Leaders (IDs 12763627, 28126340).** 18+ and 12+ year senior engineers applying to Associate SWE scored high on technical fit but represented an immediate flight risk if fast-tracked into entry-level roles. *Fix:* Gate 7 Rule 8 (`OVERQUALIFIED`) intercepts experience exceeding role seniority by $\geq$ 2 tiers and routes to **ESCALATE** for compensation alignment.
- **F3, Perfect Score, Zero Verification (ID 16186411).** A 100/100 text match on Associate SWE carried zero GitHub links and zero public repositories. *Fix:* Gate 7 Rule 4 diverts unverified profiles scoring > 90 with no live artifacts to **ESCALATE**, guarding against fabricated or AI-generated resumes.
- **F4, Premature Elimination of Non-Traditional Talent (ID 33381211).** One verified skill and a Gate 5 cosine score below 0.30 triggered premature auto-rejection despite genuine backend relevance. *Fix:* auto-rejection now requires *both* score < 50.0 *and* zero verified skills; any verified skill routes to **ESCALATE** instead.
- **F5, Dense Layout & Non-Standard Headers (SYN035, SYN042).** Gate 1 parsed text correctly, but Gate 3 missed embedded technologies (Redis, Kubernetes) inside narrative paragraphs under non-standard section headers ("Learnings & Tooling"). *Fix:* added paragraph-level sentence splitting in Gate 1 and augmented Gate 3 few-shot prompts with international CV idioms.

8 Intellectual Honesty: What Is Misleading?

Here is where the headline numbers above need an honest asterisk, stated plainly rather than buried in a footnote:

- **Extraction F1 is a clean-subset number.** The reported 1.00 F1 was measured on a clean, structured 20-candidate subset. On uncurated resumes with ambiguous date formats ("Winter 2021"), non-standard degree titles, and skills buried in narrative prose, true production performance is conservatively estimated at **0.85 to 0.90** for skills and **0.92** for experience years. Presenting 1.00 without this qualifier would overstate readiness.
- **Auto-advance is intentionally conservative (2/8 stars).** Of 8 ground-truth stars, 2 were auto-advanced and 6 escalated: 100% precision but 25% recall at that gate. This is a deliberate ultra-high-precision fast-track (≥ 85.0 score, $\geq 60\%$ must-haves, zero flags); the remaining 6 stars were safely escalated, not lost. The system's true operational value is the 84 automatic, zero-loss rejections (90.32% precision), a 42% cut in recruiter reading volume.
- **Synthetic vs. real NDCG discrepancy.** Early synthetic stress testing produced NDCG@10 = 0.8825; the real 200-pair Kaggle golden set yields 0.8033. Synthetic profiles were more polarized than reality. Conflating the two numbers would overstate production readiness.
- **Decision accuracy is structurally capped by recall bias.** 82.50% accuracy may look lower than an unconstrained classifier's claimed 90%+, but the ceiling is deliberate: 105/200 candidates are escalated by design rather than mechanically resolved, trading raw accuracy for protection of non-traditional talent.
- **Lexical baselines can win on keyword-dense JDs.** On boilerplate-heavy postings, TF-IDF's 0.9074 NDCG beat HYT's 0.6153 (Table 2) precisely because HYT penalizes unverified claims and overqualification that a naive lexical match rewards.

9 Engineering Decision Log (Selected)

- **Domain specialization.** Software Engineering + IT Infrastructure adjacency was chosen deliberately to force the system to distinguish true developers from sysadmins/technicians rather than treating all cross-domain applicants as uniform rejects.
- **Ground-truth-first calibration.** All 200 golden pairs were hand-labelled *before* scoring logic was finalized, so every gate and threshold could be measured against real precision/recall/NDCG rather than guessed.
- **Deterministic scoring over neural guessing.** LLMs are structurally forbidden from producing the final score; this eliminates drift, calibration bias, and injection-induced score manipulation by construction.
- **Structural recall bias.** False negatives are penalized $\geq 5\times$ more severely than false positives; ambiguity always resolves to **ESCALATE**, never silent rejection.
- **Immutable evidence ledger.** No dimensional score may exist without a `LedgerEntry` citing verbatim resume text and a named JD gap; Pydantic validators reject incomplete ledgers outright.
- **Two-LLM-call hard ceiling.** Exactly one extraction call (G3) and one explanation call (G8) per candidate, versus 5 to 10 calls in typical multi-agent designs, cutting token spend by roughly 70% while keeping cached re-runs free.
- **Pre-extraction PII redaction.** Fairness is architectural: demographic tokens are physically removed before Gate 3 ever sees the text, rather than relied upon via a system-prompt instruction to the LLM.
- **No hard-coded thresholds.** Every weight and cutoff loads from `config.yaml`, validated at startup (dimension weights verified to sum to 1.0), enabling reproducible calibration sweeps without touching source code.
- **Non-fatal floors for education and location.** Neither dimension can alone force a score to zero or kill a candidacy, protecting self-taught and relocation-flexible engineers from blunt disqualification.
- **Three-tier skills-taxonomy cascade.** Canonical alias lookup (`k8s` $\leftrightarrow$ `kubernetes`) $\rightarrow$ word-boundary regex guard (blocks `Java` matching inside `JavaScript`) $\rightarrow$ dense-embedding fallback gated at cosine ≥ 0.68. This combines instant determinism on acronyms with semantic coverage on genuine paraphrases.

10 Roadmap: What Next With One More Week

- **Active learning from recruiter overrides.** A nightly regression job trains on logged override events to recalibrate dimensional weights toward observed recruiter judgment.
- **Cross-encoder reranking for the mid-band (50–70).** A dedicated cross-encoder (`ms-marco-MiniLM-L-6-v2`) applied only to borderline candidates, to evaluate deep project-narrative alignment beyond bag-of-skills matching.
- **Taxonomy expansion into Data & ML Engineering.** Extend the canonical taxonomy to MLOps, PyTorch, and distributed-training vocabulary for adjacent job families.
- **Automated, compliant candidate feedback.** Use Gate 8's grounded ledger to generate rejection notices that cite verifiable skill gaps without exposing proprietary scoring weights.